Final Report: The Dual-Axis Molecule Dataset (Version 3)

A plain-language guide to how the dataset was built, what is inside it, and how to use it | build script: build_dataset_v3.py | seed 42 | sources: Tox21 and SIDER

1. What is this dataset for?

Imagine a model that has learned from thousands of normal molecules. Later it meets a molecule that looks 'unusual'. There are two very different reasons why a molecule can look unusual:

- **It is broken.** The chemistry is impossible, for example a carbon atom with five bonds (carbon can only have four). The right action is to throw it away.
- **It is new but real.** The molecule is chemically fine, it just looks different from the training data. The right action is to send it to an expert, because it might be an interesting discovery.

Most existing methods give one single 'how unusual is it?' score, so they cannot tell these two cases apart. This project tests a method that uses two scores: one for **validity** (is it chemically possible?) and one for **novelty** (is it unlike the training data?).

This dataset is the test bench for that idea. It contains normal molecules, new-but-real molecules and broken molecules, and every molecule is clearly labeled with which kind it is.

Scope of this work: only the dataset was built, so the next team member can explore it (EDA). No model has been trained yet.

2. The dataset at a glance

- **Two source datasets:** Tox21 (7,831 molecules with 12 toxicity lab results each) and SIDER (1,427 marketed drugs). Both come from the public DeepChem/MoleculeNet files.
- **After cleaning:** 7,545 Tox21 molecules and 1,303 SIDER molecules.
- **Ten sets** of molecular graphs, roughly 10,000 graphs in total: 3 normal sets, 4 new-but-real sets and 3 broken sets.

- **Every broken molecule has a normal 'twin'**: the original molecule it was made from, in the same split.
- **Fully reproducible:** fixed random seed, recorded software versions, and fingerprints (SHA-256 hashes) of every input and output file.
- **Nothing is ever overwritten:** the build script refuses to write into a folder that already has files.

3. The three kinds of molecules

1) Normal (called in-distribution, or ID). Tox21 molecules used to train and test the model. Sets: id_train, id_val, id_test.

2) New but real (called novelty). Valid molecules that the model has not seen the like of. There are two flavors:

- *New structure:* Tox21 molecules whose ring skeleton (scaffold) never appears in any normal set. Sets: novel_scaffold_val, novel_scaffold_test.
- *New source:* drug molecules from a different dataset (SIDER). Sets: novel_dataset_val, novel_dataset_test.

3) Broken (called corrupted). Normal molecules that were deliberately damaged so they break a chemistry rule. Sets: corrupt_train, corrupt_val, corrupt_test.

The 'val' sets are for tuning a model. The 'test' sets are for the final reported results. Never tune on a test set.

4. How the dataset was built, step by step

Step 0: Setup

We fixed the random seed (42) so the same inputs always give the same dataset. We recorded the software versions and took a fingerprint of the raw files. The script will not overwrite old results.

Step 1: Load the raw data

We loaded Tox21 (the normal data) and SIDER (the source of new-but-real drugs). For Tox21 we also checked that every toxicity result is 0, 1 or missing. For SIDER we only use the molecule structures, not its side-effect columns.

Step 2: Clean the molecules

Molecules are written as text called SMILES (for example CCO is ethanol). We cleaned them in four ways:

- **Read each one with chemistry software (RDKit).** 8 Tox21 entries could not be read. All 8 were aluminium compounds with unusual bonding, so they were dropped.
- **Keep the largest piece.** Many entries are salts or mixtures made of several pieces. We kept only the biggest piece (244 Tox21 and 182 SIDER entries had more than one piece).
- **Size filter.** We kept molecules with 3 to 100 heavy (non-hydrogen) atoms. Dropped for being too small / too large: 31 / 7 in Tox21 and 36 / 42 in SIDER.
- **Remove duplicates.** The same molecule can appear several times (for example as different 3D versions or salts). We compared molecules in a standard written form that ignores 3D shape, and merged 240 duplicate rows in Tox21 and 46 in SIDER. When duplicates disagreed on a toxicity result (95 cases), that one result was set to missing rather than guessed.

Result: 7,831 Tox21 rows became 7,545 molecules, and 1,427 SIDER rows became 1,303. Every removed row is recorded in dropped_rows.csv with the reason.

Step 3: Scaffolds (the ring skeleton)

A scaffold is what is left of a molecule when you remove all side branches and keep only the rings and the links between them. Two molecules with the same ring skeleton belong to the same 'scaffold family'. We use this to define 'new structure' fairly: a molecule is structurally new if its whole scaffold family is new.

Some molecules have no rings at all ('ring-free'). They have no scaffold, so they are handled separately and are never treated as a 'new structure'.

Step 4: Remove overlap between Tox21 and SIDER

880 of the 1,303 SIDER molecules also appear in Tox21. We removed them, because a 'new' molecule that is actually in the training data would be a false test. 423 SIDER molecules remain.

Step 5: Split the Tox21 data

- **First we set aside the new-structure pool.** About 10% of the Tox21 molecules (755 molecules from 407 scaffold families) were taken out as whole scaffold families. They are never used for training, and only ring-containing molecules are eligible.

- **Then we split the remaining 6,790 molecules** into train, validation and test, about 80% / 10% / 10%:
  - Ring-containing molecules are split by scaffold family, so the scaffolds in validation and test are not seen in training.
  - Ring-free molecules have no scaffold, so they are split randomly with the same fractions. This keeps all three splits looking alike.

The ring-free rule was added after a quality check on an earlier build found that nearly all ring-free molecules had ended up in validation and test (see section 7).

Step 6: Build the new-but-real sets

The new-structure pool and the SIDER molecules were each cut into a validation half and a test half, again by scaffold family, so no family appears in both halves. We then tried to make each set the same size as its matching normal set. There were not enough molecules, so each set keeps everything it has and is marked size_matched = False.

- novel_scaffold_val: 408 | novel_scaffold_test: 347
- novel_dataset_val: 225 | novel_dataset_test: 198

Because these new-but-real sets are larger or more complex than the normal ones, we also built a table of 'matched controls': for each new-but-real molecule, a normal molecule of the same split with nearly the same size (within 2 atoms and 2 bonds) and the same ring-free status. This lets you compare fairly without molecule size doing the work.

Step 7: Build the broken molecules

Each broken molecule is a copy of a normal molecule from the same split (its twin) with 1 to 3 changes. There are three kinds of damage (called families):

- **Family A, stronger bond:** a single bond becomes a double bond, or a double becomes a triple, on an atom that is already full.
- **Family B, moved bond:** one end of a bond is moved onto an atom that is already full. The number of bonds stays the same and the molecule stays in one piece.
- **Family C, wrong atom:** a carbon atom is swapped for oxygen. This family exists only in the test set, so we can see how a model copes with a kind of damage it never saw in training.

Every change is only accepted if it creates a real valence error (an atom with too many bonds), and the finished molecule is then re-checked so that this is its only chemistry problem. If a small molecule cannot take the planned number of changes, it gets fewer.

Sizes: about 20% of the normal set size for training, and about 80% for validation and test.

Step 8: Turn molecules into graphs

A molecule is a graph: atoms are the nodes and bonds are the links. The same code builds every graph in every set, including the broken ones.

- **Each atom has 30 numbers:** which element, how many connections, its charge, how many hydrogens it carries, whether it is aromatic, and whether it is in a ring.
- **Each bond has 6 numbers:** its type (single, double, triple, aromatic, other) and whether it is in a ring.

We deliberately left out some features (such as how the atom is hybridized, its exact valence, and its 3D handedness) because for broken molecules they would give away the edit.

Step 9: Attach the labels

The 12 Tox21 toxicity results (0, 1 or missing) are attached to the normal sets and the new-structure sets. SIDER molecules and broken molecules have all-missing labels, because they do not have Tox21 results. For a broken molecule, use its twin to look up the original labels.

Step 10: Quality checks (audits)

Before saving, the script runs checks and refuses to call the build PASSED if a serious one fails:

- **Leakage:** no molecule appears in two sets, and no ring-containing scaffold appears across normal splits or between normal and new-structure sets.
- **Twins:** every broken molecule has its twin in the same split, and each twin is used only once.
- **Corruption:** every broken molecule fails the valence rule and nothing else, and the number of edits matches the record.
- **Format:** all graphs have the right shapes, no missing numbers, and symmetric links.
- **Balance:** a new check warns if the normal splits do not look alike in ring-free share.
- **Shortcut audit (explained in section 7).**

On the earlier build, we also checked the results independently, outside the script, from the saved files. That check is described in section 7 and must be repeated on the final build.

Step 11: Save everything

The script writes the graphs, tables, split lists, audit results, a report, a data card and a config file with fingerprints. The next section lists every file.

5. Every file in the output folder

Top level

- BUILD_REPORT.md: the run summary. Its first line says PASSED or FAILED. Read it first.
- DATA_CARD.md: a short description to keep with the data.
- config.json: the exact settings, software versions, raw file fingerprints, and fingerprints of every output file.
- feature_schema.json: the names of the 30 atom features, 6 bond features and 12 labels.
- AUDIT_FAILED.txt: appears only if a serious check failed. If you see it, do not use the dataset.

graphs/ (10 files, one per set)

- id_train.pt, id_val.pt, id_test.pt, novel_scaffold_val.pt, novel_scaffold_test.pt, novel_dataset_val.pt, novel_dataset_test.pt, corrupt_train.pt, corrupt_val.pt, corrupt_test.pt.
- Each file is a list of graphs. Each graph has: uid (its name), x (atom features), edge_index (which atoms are linked), edge_attr (bond features) and y (the 12 labels).

tables/

- master_table.csv: the main file for exploration. One row per graph: set, split, kind, SMILES, scaffold, size, ring-free flag, twin, damage family, number of edits, and the 12 labels.
- eligible_pairs.csv: each broken molecule linked to its normal twin.
- matched_controls.csv: each new-but-real molecule linked to a size-matched normal molecule.
- dropped_rows.csv: every molecule removed anywhere, with the stage and the reason.

splits/

- split_ids.json: the list of molecule IDs in each of the 10 sets.

audits/

- audit_results.json: the results of every check in machine-readable form.

logs/

- build.log: the full record of the run.

6. How to open the data

```
import torch, pandas as pd
graphs = torch.load('processed_v3/graphs/id_train.pt', weights_only=True)
master = pd.read_csv('processed_v3/tables/master_table.csv')

# optional: convert one graph for PyTorch Geometric
from torch_geometric.data import Data
d = graphs[0]
data = Data(x=d['x'], edge_index=d['edge_index'], edge_attr=d['edge_attr'],
y=d['y'])
```

Join the graphs to the master table using the uid.

7. Quality checks, a problem we found, and the shortcut audit

What the independent check confirmed (on the earlier build):

- All output file fingerprints matched.
- The tables matched the graphs exactly.
- No molecule or scaffold leaked between sets.
- Every twin was in the right split and used once.
- Every graph was connected.
- A separate valence rule found 0% broken molecules among the normal sets, and about 98% among families A and B.

A problem we found and fixed. In the earlier build, about 53% of the normal validation and test molecules had no rings, against only 18% in training. The scaffold split had pushed almost all ring-free molecules into validation and test. That made 'normal' test data differ from training data, and it exaggerated how easy novelty was to detect. The fix is the random split for ring-free molecules (Step 5), plus matching on ring-free status in the control table (Step 6), plus a new warning in the audit.

The shortcut audit, in plain words. We trained a very simple model that only counts things (how many atoms, bonds, and of which elements). If this counting model can already separate two groups, then the groups differ in an easy way and a fancy model

might just be using that cue instead of real understanding. A score of 0.5 means no cue; 1.0 means the cue is perfect. Findings:

- **Family A:** no countable difference at all. This is the cleanest test of validity.
- **Family B:** reduced, because bonds are moved rather than added, but the number of connections per atom still changes a little.
- **Family C:** swapping an element necessarily changes the element counts. This cannot be removed, so report it separately.
- **New-but-real sets:** they are larger and chemically a bit different from normal molecules. Use the matched controls to compare fairly.

The lesson: always report results per damage family, and always include the simple counting model as a baseline.

8. Known limits

- Only valence errors are used as 'broken'. Other kinds of invalid molecules are not covered, and the idea has only been tested on chemistry.
- Family B and C keep some countable difference; only family A is clean.
- Small molecules cannot always take 3 edits, so the number of edits tends to be higher for larger molecules.
- SIDER gives only 423 usable molecules, so results on it have wide uncertainty. About a third of its ring-containing molecules share a scaffold with Tox21 training data, so it is a different-source test, not a new-structure test.
- The new-but-real sets are smaller than the normal sets (size_matched = False). Account for this when comparing.
- Some toxicity tasks have few positive examples in validation and test, so per-task scores can be unreliable.
- Broken molecules have no labels of their own; use the twin.

9. Suggested first steps for exploration (EDA)

1. Open BUILD_REPORT.md and confirm it says PASSED. Read every warning.
2. Plot the number of atoms, bonds and the element mix for every set.
3. Compare normal vs new-but-real using matched_controls.csv, not the raw normal set.
4. For each damage family, compare twins using eligible_pairs.csv.

5. Check the positive rate and the missing rate of each of the 12 labels in each set.
6. Look at scaffold family sizes and how far the new-structure sets sit from training.
7. Build a counting-only baseline first, and compare every later model against it.

10. Glossary

- **SMILES:** a text way to write a molecule.
- **Graph:** atoms as dots, bonds as lines between them.
- **Scaffold:** the ring skeleton of a molecule with side branches removed.
- **Ring-free molecule:** a molecule with no rings, so it has no scaffold.
- **Valence:** how many bonds an atom is allowed to have (carbon: 4, oxygen: 2).
- **In-distribution (ID):** looks like the training data.
- **Novelty / OOD (out-of-distribution):** valid but unlike the training data.
- **Corrupted:** deliberately damaged so it breaks a valence rule.
- **Twin:** the normal molecule that a broken molecule was copied from.
- **AUROC:** a score from 0 to 1 for how well two groups can be told apart. 0.5 means not at all.
- **Shortcut:** an easy cue (such as molecule size) that lets a model separate groups without understanding the real difference.

11. Numbers to fill in from the final build

The cleaning counts, the novelty set sizes, the 755-molecule pool and the 423 SIDER molecules in this report do not depend on the final run. The numbers below do, so copy them from processed_v3/BUILD_REPORT.md once the version 3 build has finished:

- id_train / id_val / id_test sizes: _____ / _____ / _____ (expected about 5,430 / 680 / 680)
- Ring-free share in id_train / id_val / id_test: _____ / _____ / _____ (expected to be similar)
- corrupt_train / corrupt_val / corrupt_test sizes: _____ / _____ / _____
- Audit status and warning count: _____
- Count-only AUROC for families A / B / C: _____ / _____ / _____
- Count-only AUROC for each new-but-real set, with and without matching: _____